

8.5

$$N \left(\frac{d^2 u}{dy^2} \right) = \frac{dP}{dx} = \text{const}$$

a). $N \left(\frac{d^2 u}{dy^2} \right) = \frac{dP}{dx} = \text{const}$ 이다

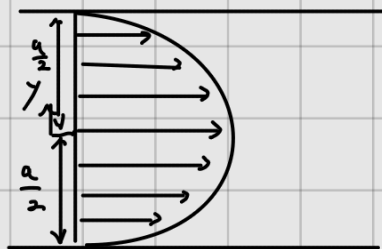
$$\frac{d^2 u}{dy^2} = \frac{1}{N} \frac{dP}{dx}$$

$$\frac{du}{dy} = \frac{1}{N} \frac{dP}{dx} y + C_1$$

$$u = \frac{1}{2N} \frac{dP}{dx} y^2 + C_1 y + C_2 \quad \text{이때 } y = \frac{a}{2}, u = 0 \quad (\text{No slip condition})$$

$$y = \frac{a}{2}, u = 0$$

$$\begin{aligned} \therefore u &= \frac{1}{2N} \frac{dP}{dx} y^2 - \frac{1}{2N} \frac{dP}{dx} \left(\frac{a}{2} \right)^2 \\ &= \frac{3}{2} \times \left(-\frac{a^2}{12N} \left(\frac{dP}{dx} \right) \right) \left(1 - \frac{4}{a^2} y^2 \right) \\ &= \frac{3}{2} u_m \left(1 - \frac{y^2}{(a/2)^2} \right) \end{aligned}$$



b) $D_h = \frac{4A}{P}$ 3 개지나... P : perimeter

$$= \frac{4(ab)}{2a+2b}$$

$b \rightarrow \infty$ 이면

$$= \frac{4a}{\frac{2a}{b} + 2} = 2a$$

$$\therefore f = \frac{-\frac{dP}{dx} D_h}{8\mu u_m^2/2}$$

c). a)에서 얻은 속도분포를 활용하면

$$\begin{aligned} f &= -\frac{dP}{dx} \frac{2D_h}{8\mu u_m^2} \\ &= -\frac{dP}{dx} \frac{2D_h}{-\frac{a^2}{12N} \left(\frac{dP}{dx} \right) 8\mu u_m} \end{aligned}$$

$$= \frac{24 \nu D_h^2}{a^2 u_m D_h}$$

$$= \frac{24 D_h^2}{a^2 Re_{Dh}} = \frac{96}{Re_{Dh}} \quad \therefore C = 96$$

d) $u_m = -\frac{a^2}{12N} \left(\frac{dP}{dx} \right)$

$$= \frac{a^2}{12N} \frac{\Delta P}{\Delta x}$$

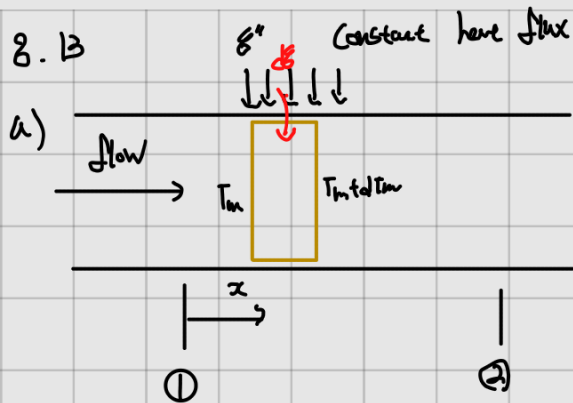
$$= \frac{(0.005)^2}{12 \times 6 \times 10^{-7}} \times \frac{3.45}{0.2}$$

$$= 2.12 \text{ m/s}$$

$$\therefore Re_{Dh} = \frac{\rho u_m D_h}{\mu} = \frac{2.12 \times 2 \times 0.005}{15.89 \times 10^{-6}} = 1332$$

$\therefore$ laminar flow $\therefore \frac{L}{d} = 0.05 Re_{Dh}$

$$L = 0.05 \times 1332 \times 2 \times 0.005 = 0.67 \text{ m}$$



CV를

$$\frac{dT}{dx} = \dot{q}'' + m \dot{h}_{in} - m \dot{h}_{out}$$

$$0 = \dot{q}'' + m (C_p T_m - C_p (T_m + \Delta T_m))$$

$$\therefore \dot{q}'' = m C_p \Delta T_m$$

$$\text{이때 } \dot{q}'' = \dot{q}'' p dx$$

$$\therefore \dot{q}'' p dx = m C_p dT_m$$

$\dot{q}'' = \text{const}$ 이므로 + (constant properties

$$T_m(x) = \frac{\dot{q}'' p}{m C_p} x + C_1 \quad (x=0, T_m = T_{m,i})$$

$$\therefore T_m(x) = \frac{\dot{q}'' p}{m C_p} x + T_{m,i}$$

$\therefore T_m$ 은 직선 분포를 보인다.

CV로 대를 연관한 이므로

$$\dot{q}'' = h(T_s - T_m)$$

Case A: thermally fully developed.

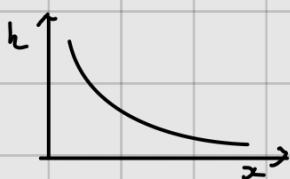
h 가 바뀌지 않는다.

$$\therefore \frac{\dot{q}''}{h} = \text{const}$$

$$T_m + \frac{\dot{q}''}{h} = T_s$$

$\therefore T_s$ 는 T_m 과 $\frac{\dot{q}''}{h}$ 만큼 차이 나는 극선분

Case B: entrance



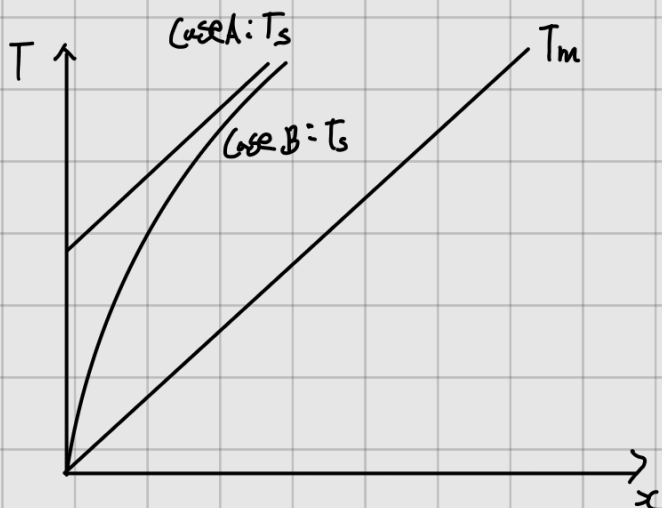
의 형태를 가지므로 $T_m + \frac{\dot{q}''}{h} = T_s$ 이다

T_s 는 T_m 과 1영의 차이만큼 가질

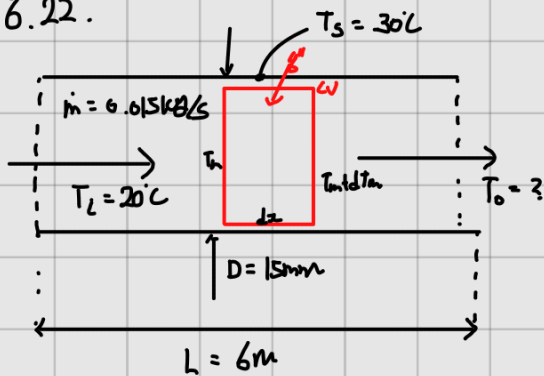
b) Case B의 경우 T_s, T_m 의 차이가 1영으로 줄어들어서 그 수직분이

Case A의 차이가 되므로 Case A의 열전달이 더 크다.

$$\therefore T_{m,2,A} > T_{m,2,B}$$



8.22.



Constant Surface Temperature, Conv. Properties

a) Conv. coef

$$\frac{dE}{dx} = Q - \dot{m}c_p \frac{dT_m}{dx}$$

$$0 = \dot{m}c_p(T_o - T_i) + \dot{m}c_p(T_m - T_o)$$

$$\frac{dT_m}{dx} = \frac{\dot{m}c_p(T_o - T_m)}{\dot{m}c_p}$$

$$\frac{dT_m}{dx} = \frac{\pi D h (T_s - T_m)}{\dot{m}c_p} \quad \text{let } \theta = T_s - T_m$$

$$\frac{d\theta}{dx} = -\frac{\pi D h}{\dot{m}c_p} \theta$$

$$\int_{\theta_i}^{\theta_o} \frac{d\theta}{\theta} = \int_0^L -\frac{\pi D h}{\dot{m}c_p} dx$$

$$\ln \frac{\theta_o}{\theta_i} = -\frac{\pi D h L}{\dot{m}c_p} \quad \dots (1)$$

$$\dot{m} = \rho A V$$

$$V = \frac{\dot{m}}{\rho A}$$

$$\therefore Re_D = \frac{VD}{\nu} = \frac{\dot{m}D}{AN} = \frac{4\dot{m}}{\pi D N} = 1487$$

$\therefore$ laminar

$$\frac{z}{D} = 0.05 Re_D Pr = 432$$

$$\therefore z = 432 D = 6.5 \text{ m}$$

$\therefore$ 전열의 entrance region

$$\text{이때의 } \overline{Nu_D} = \frac{3.36}{\tanh(2.264 Gz_D^{-1/3} + 1.7 Gz_D^{-2/3})} + 0.0499 Gz_D \tanh(Gz_D^{-1})$$

$$\frac{3.36}{\tanh(2.264 (2.432)^{-1/3} + 1.7 (2.432)^{-2/3})} + 0.0499 (2.432) \tanh(2.432^{-1})$$

$$\therefore \bar{h} = 203 \text{ W/m}^2 \cdot \text{K}$$

(1)에 대입

$$\ln \frac{\theta_o}{\theta_i} = -\frac{\pi D h L}{\dot{m}c_p}$$

$$\theta_o = \theta_i \exp\left(-\frac{\pi D h L}{\dot{m}c_p}\right)$$

$$= 10 \exp\left(-\frac{203 \times \pi \times 0.015 \times 6}{0.015 \times 4200}\right)$$

$$\theta_o = 4.02$$

$$\therefore T_{m,o} = 26$$

$$\therefore \dot{Q} = \dot{m}c_p(T_o - T_i)$$

$$= 378 \text{ W}$$

b) for non fluids

$$k = 0.105 \text{ W/m}\cdot\text{K} \quad \rho = 1146 \text{ kg/m}^3 \quad c_p = 3.587 \text{ kJ/kg}\cdot\text{K} \quad \mu = 962 \times 10^{-6} \text{ N}\cdot\text{s/m}^2 \quad \alpha = 171 \times 10^{-9} \text{ m}^2/\text{s}$$

$$\dot{m} = \rho A V = 0.015 \text{ kg/s}$$

$$\therefore V = \frac{0.015}{\rho A} = 0.074 \text{ m/s}$$

$$\therefore Re_D = \frac{VD}{\mu} = 1322 \quad (\text{laminar})$$

$$Pr = \frac{V}{\alpha} = 4.91$$

$$\frac{x}{D} = 0.05 Re_D Pr$$

$$x = 4.87 \text{ m}$$

$$Gz_D = \left(\frac{D}{x}\right) Re_D Pr = \frac{15 \times 10^{-3}}{4.87} \times (1322 \times 4.91) = 19.97$$

$$\text{Combined entry eqn } \frac{Nu_D}{Nu_{\infty}} = \frac{3.36}{\tanh(2.264 Gz_D^{-1/3} + 1.7 Gz_D^{-2/3})} + \frac{0.0499 Gz_D \tanh(Gz_D^{-1})}{\tanh(2.432 Pr^{1/6} Gz_D^{1/6})}$$

$$\therefore \bar{h} = 220 \text{ W/m}^2\cdot\text{K}$$

$$\therefore \ln \frac{\theta_o}{\theta_i} = -h \frac{620}{\dot{m} c_p}$$

$$= 10 \exp \left(-220 \times 620 \times 15 \times 10^{-3} \div 0.015 \div 3500 \right)$$

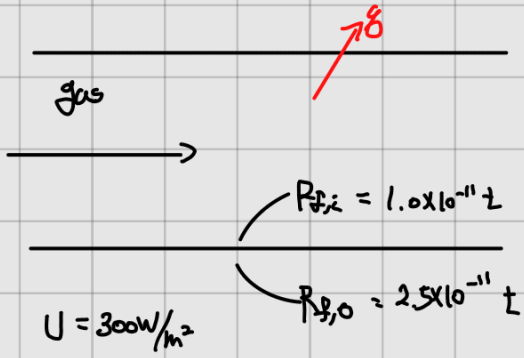
$$\theta_o = 3.147$$

$$\therefore T_{m,o} = 26.8^\circ\text{C}$$

$$\therefore \dot{Q} = \dot{m} c_p (T_{m,o} - T_{m,i}) = 365 \text{ W}$$

11.2

water

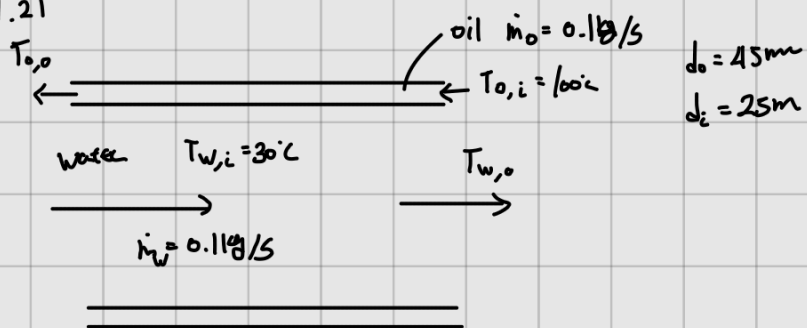


$$\frac{1}{U_{\text{clean}}} = \frac{1}{U} + R_{f,i} + R_{f,o}$$

$$\frac{1}{0.15 \times 300} = \frac{1}{300} + a_i t + a_o t$$

$$\therefore t = \frac{\frac{1}{300 \times 0.15} - \frac{1}{300}}{a_i + a_o} = 31746031 \text{ s}$$

11.21



a) If $T_{o,o} = 60^\circ\text{C}$ $Q = ?$, $T_{w,o} = ?$

$$Q = \dot{m} c_{p,o} T_{o,o} - \dot{m} c_{p,o} T_{o,i}$$

$$= \dot{m} c_{p,o} (T_{o,o} - T_{o,i})$$

$$= (0.1)(1900)(-40)$$

$$= -7600 \text{ J from oil}$$

then for water

$$Q = \dot{m} c_{p,w} (T_{w,o} - T_{w,i})$$

$$7600 = (0.1)(4200)(T_{w,o} - 30)$$

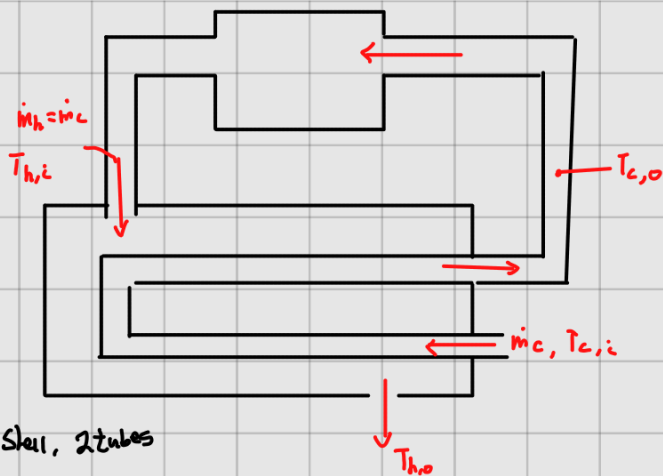
$$\therefore T_w = 48.1^\circ\text{C}$$

b) $Q = \pi D_i L U \Delta T_{lm} = 7600$

$$\Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln \frac{\Delta T_1}{\Delta T_2}} = \frac{51.9 - 30}{\ln \frac{51.9}{30}} = 39.95$$

$$\therefore L = \frac{Q}{\pi D_i L U \Delta T_{lm}} = \frac{7600}{\pi (25 \times 10^{-3})(60)(39.95)} = 40.4 \text{ m}$$

11.24.



$$T_{c,i} = 5^\circ\text{C}$$

$$T_{h,i} = 170^\circ\text{C}$$

$$\rho = 1030 \text{ kg/m}^3$$

$$C_p = 3860 \text{ J/kg}\cdot\text{K}$$

1 Shell, 2 tubes

a) $\dot{m} = 5 \text{ kg/s}$, $U = 2000 \text{ W/m}^2\cdot\text{K}$, $\varepsilon = 0.5$, $A = ?$, $T_o = ?$, $\xi = ?$

$$C_c = \dot{m} C_p = 1930 \text{ kJ/K}$$

$$\therefore C_r = 1$$

$$C_h = \dot{m} C_p = "$$

$$\therefore \varepsilon = \frac{2/\varepsilon - (H C_p)}{(H C_p)^{1/2}} = 1.214$$

$$\therefore \text{NTU} = -(H C_p)^{-1/2} \ln\left(\frac{\varepsilon - 1}{\varepsilon + 1}\right)$$

$$\frac{UA}{C} = 1.2468$$

$$\therefore A = \frac{1930}{2000} \times 1.2468 = 12.03 \text{ m}^2$$

$$\xi = \varepsilon C (T_{h,i} - T_{c,i})$$

$$= 627250 \text{ W}$$

$$\therefore \xi = C (T_{c,o} - T_{c,i})$$

$$T_{c,o} = 37.5^\circ\text{C}$$

$$\xi = C (T_{h,i} - T_{h,o})$$

$$T_{h,o} = 37.5^\circ\text{C}$$

b) $\eta = 0.9$, $C_{\text{ng}} = \$ 0.02/\text{MJ}$, Annual Savings = ?

by installation we get heat recovery rate $\xi = 627250 \text{ J}$

$$\therefore \Delta E = Q_{\text{lost}} = \xi \times 3600 \times 24 \times 365 = 1.978 \times 10^{13} \text{ J}$$

$$= 1.978 \times 10^7 \text{ MJ}$$

$$\therefore \text{Annual Savings} = \frac{\Delta E \times 0.02/\text{MJ}}{0.9} = \underline{\underline{439576 \$}}$$